

Day-Ahead Power Price Forecasting (DE-LU & GB) - Executive Summary

Key Takeaways: Across expanding-window walk-forward evaluations, Catboost is the statistically dominant model in DE-LU with an out-of-sample MAE of 20.58 EUR/MWh (vs Naive-24h 27.98 EUR/MWh). In GB, Lightgbm achieved an out-of-sample MAE of 16.18 GBP/MWh (vs Naive-24h 21.67 GBP/MWh). Diebold-Mariano tests confirm that machine learning and hybrid architectures statistically significantly outperform traditional heuristics.

Germany-Luxembourg (DE-LU) Performance

Model	MAE (EUR)	RMSE (EUR)	sMAPE (%)	DM p-val	DM Winner
Catboost	20.58	28.19	36.5%	0.0000	YES
Xgboost	21.84	29.35	37.4%	0.0003	YES
Lightgbm	21.93	30.16	37.4%	0.0005	YES
Hybrid	24.04	33.23	40.9%	0.0280	YES
Naive-24H	27.98	41.48	48.7%	1.0000	-
Seasonal-Naive-Avg	29.62	39.94	47.0%	0.1876	-
Lstm	33.77	45.68	46.1%	0.0019	-
Naive-Week	34.81	48.32	54.6%	0.0067	-
Climatological-Mean	88.39	97.32	74.6%	0.0000	-

Great Britain (GB) Performance

Model	MAE (GBP)	RMSE (GBP)	sMAPE (%)	DM p-val	DM Winner
Lightgbm	16.18	26.93	23.9%	0.0000	YES
Catboost	16.46	27.07	24.2%	0.0000	YES
Xgboost	16.56	27.67	24.1%	0.0000	YES
Hybrid	19.64	31.82	27.5%	0.0530	-
Seasonal-Naive-Avg	21.60	34.55	30.2%	0.9458	-
Naive-24H	21.67	36.02	33.5%	1.0000	-
Lstm	25.31	38.69	35.1%	0.0093	-
Naive-Week	26.86	43.80	39.5%	0.0090	-
Climatological-Mean	41.23	49.96	44.2%	0.0000	-